

Module 6: Application Layer

Computer Networks

Q1. Explain DNS and its working.

DNS (Domain Name System) is a hierarchical, distributed naming system that translates human-readable domain names (e.g., `www.google.com`) into IP addresses (e.g., `142.250.64.100`) that computers use to communicate.

Why DNS is needed: Humans remember domain names easily, but computers route traffic using IP addresses. DNS acts as the “phone book” of the Internet.

DNS Hierarchy:

- **Root DNS Servers:** Top of the hierarchy; know the addresses of TLD servers.
- **TLD (Top-Level Domain) Servers:** Handle extensions like `.com`, `.org`, `.in`.
- **Authoritative DNS Servers:** Contain actual IP records for specific domains.
- **Local DNS Resolver:** The client-side resolver that queries on behalf of the user.

Working (Step-by-step):

1. User types `www.example.com` in the browser.
2. The **local DNS resolver** checks its cache. If found, returns the IP directly.
3. If not cached, the resolver queries the **Root DNS server**.
4. The Root server directs it to the **TLD server** (e.g., `.com`).
5. The TLD server directs it to the **Authoritative DNS server** for `example.com`.
6. The Authoritative server returns the IP address.
7. The resolver caches the result and returns it to the browser.
8. The browser connects to the server using the IP address.

DNS uses UDP on port 53 for most queries (fast, low overhead).

Q2. Explain HTTP and HTTPS.

HTTP (HyperText Transfer Protocol) is an application-layer protocol used for transferring web pages and resources between a web server and a client (browser).

- Uses **port 80**.
- Follows a **request-response** model: the client sends an HTTP request; the server responds with the resource and a status code.
- **Stateless:** Each request is independent; the server does not remember previous requests.
- Common HTTP methods: `GET`, `POST`, `PUT`, `DELETE`.

HTTPS (HTTP Secure) is the secure version of HTTP that encrypts communication using

TLS (Transport Layer Security).

- Uses **port 443**.
- Provides **confidentiality** (encryption), **integrity** (data not tampered), and **authentication** (server identity verified via SSL/TLS certificate).
- Essential for online banking, e-commerce, login pages.

Comparison:

Feature	HTTP	HTTPS
Port	80	443
Security	No encryption	Encrypted (TLS)
Certificate	Not required	SSL/TLS certificate required
Speed	Slightly faster	Slightly slower (encryption overhead)
Use Case	General browsing	Sensitive transactions

Q3. Explain FTP architecture and working.

FTP (File Transfer Protocol) is an application-layer protocol used for transferring files between a client and a server over a TCP/IP network.

Architecture:

FTP uses a **dual-connection** model:

- **Control Connection (Port 21):** Used to send commands (e.g., LIST, RETR, STOR) and receive responses. Remains open for the entire session.
- **Data Connection (Port 20):** Used for the actual transfer of files or directory listings. Opened and closed for each transfer.

FTP Modes:

- **Active Mode:** The server initiates the data connection to the client.
- **Passive Mode:** The client initiates both connections (used when client is behind a firewall).

Working (Step-by-step):

1. Client connects to server on **port 21** (control connection).
2. Client authenticates with a username and password.
3. Client sends commands (e.g., download a file with RETR filename).
4. A separate **data connection on port 20** is opened for file transfer.
5. File is transferred; data connection is closed.
6. Client may continue with more commands or disconnect.

Note: FTP transfers data in **plaintext** (not secure). **FTPS** and **SFTP** are secure alternatives.

Q4. Explain SMTP, POP3, and IMAP.

These are the three core protocols used for **email communication**.

1. SMTP (Simple Mail Transfer Protocol)

- Used for **sending** emails from client to server and between mail servers.
- Operates on **port 25** (or 587 for submission).
- **Push protocol** — the sender pushes the mail to the server.
- Works with MTA (Mail Transfer Agent) to route mail to the recipient's server.

2. POP3 (Post Office Protocol version 3)

- Used for **retrieving** emails from the server to the client.
- Operates on **port 110** (995 for secure POP3S).
- **Downloads and deletes** emails from the server by default.
- Suitable for single-device access; offline reading supported.

3. IMAP (Internet Message Access Protocol)

- Also used for **retrieving** emails, but keeps them on the server.
- Operates on **port 143** (993 for secure IMAPS).
- Supports **multi-device access** — emails remain synced across devices.
- Supports folders, flags, and server-side search.

Comparison:

Feature	SMTP	POP3	IMAP
Purpose	Send mail	Receive mail	Receive mail
Port	25 / 587	110 / 995	143 / 993
Stores on Server	Yes (relay)	No (downloads)	Yes
Multi-device	—	No	Yes

Q5. What is Telnet? Explain remote login.

Telnet (Telecommunication Network) is an application-layer protocol that allows a user to **remotely log in** to another computer over a TCP/IP network and interact with it via a command-line interface.

- Operates on **port 23**.
- Uses a **client-server model**: the Telnet client connects to the Telnet server on the remote machine.
- Once connected, the user can execute commands on the remote system as if physically present.

Remote Login Process:

1. The Telnet client establishes a **TCP connection** to the remote host on port 23.
2. The remote server presents a **login prompt**.

3. The user enters credentials (username and password).
4. After authentication, the user gets a **command shell** on the remote machine.
5. Commands typed by the user are sent to the remote system and executed; output is returned to the client.
6. The session ends when the user logs out or the connection is closed.

Limitation: Telnet transmits all data including passwords in **plaintext**, making it highly insecure. It has largely been replaced by **SSH**.

Q6. What is SSH? How is it different from Telnet?

SSH (Secure Shell) is a cryptographic network protocol that provides a **secure, encrypted channel** for remote login, command execution, and file transfer over an unsecured network.

- Operates on **port 22**.
- Encrypts all data using **public-key cryptography** and symmetric encryption.
- Provides **authentication, confidentiality, and integrity**.
- Supports additional features: **SCP** (secure copy), **SFTP** (secure file transfer), and **port forwarding/tunneling**.

SSH vs. Telnet:

Feature	SSH	Telnet
Port	22	23
Security	Encrypted (TLS/PKI)	No encryption (plaintext)
Authentication	Password + Public Key	Password only
Data Transfer	Secure (SCP/SFTP)	Not supported
Usage Today	Widely used	Mostly obsolete
Integrity Check	Yes	No

In summary: SSH is the secure replacement for Telnet, offering all the same remote access functionality with strong encryption and authentication.

Q7. Explain NFS and SMB file-sharing protocols.

NFS (Network File System) and **SMB (Server Message Block)** are two popular protocols that allow users to access files on a remote server as if they were on a local drive.

1. NFS (Network File System)

- Developed by **Sun Microsystems**; primarily used in **Unix/Linux** environments.
- Allows a server to **export** a directory, which clients can **mount** as a local filesystem.
- Operates over **TCP/UDP** (typically port 2049).
- Transparent to applications — programs do not need modification to use network files.
- Versions: NFSv3 (UDP-based), NFSv4 (TCP, stateful, better security).

2. SMB (Server Message Block)

- Developed by **IBM**, popularized by **Microsoft Windows** (also called **CIFS**).
- Used for sharing files, printers, and serial ports over a network.
- Operates over **TCP port 445**.
- **Samba** is an open-source implementation of SMB for Linux/Unix systems.
- Supports authentication, access control, and file locking.

Comparison:

Feature	NFS	SMB
Origin	Sun Microsystems	IBM / Microsoft
Primary OS	Unix / Linux	Windows
Port	2049	445
Use Case	Server-to-server	Office file sharing
Authentication	Host-based (v3), Kerberos (v4)	Username/password

Q8. Explain the role of a domain name server.

A **Domain Name Server (DNS server)** is a specialized server that stores a database of domain names and their corresponding IP addresses, and answers queries to translate one into the other.

Roles of a DNS Server:

- **Name Resolution:** The primary role — translating human-readable domain names (e.g., `www.example.com`) into numeric IP addresses (e.g., `93.184.216.34`).
- **Reverse Lookup:** Translating an IP address back to a domain name (used in logging and security tools).
- **Caching:** DNS servers cache responses for a period defined by the **TTL (Time To Live)** value, reducing repeated lookup overhead and speeding up resolution.
- **Load Distribution:** A single domain can map to multiple IP addresses. DNS can distribute client requests across multiple servers (round-robin DNS).
- **Mail Routing:** DNS stores **MX (Mail Exchange) records** to direct email to the correct mail server for a domain.
- **Service Discovery:** **SRV records** in DNS help clients discover services (e.g., VoIP, XMPP) associated with a domain.
- **Security (DNSSEC):** DNS Security Extensions (DNSSEC) add digital signatures to DNS records to protect against spoofing and cache poisoning attacks.

Types of DNS Records:

Record Type	Purpose
A	Maps domain to IPv4 address
AAAA	Maps domain to IPv6 address
MX	Specifies mail server
CNAME	Alias for another domain name
NS	Specifies authoritative name server
PTR	Reverse lookup (IP to name)
TXT	Stores text info (SPF, DKIM, etc.)

Q9. Explain the Simple Network Management Protocol (SNMP).

SNMP (Simple Network Management Protocol) is an application-layer protocol used for **monitoring and managing network devices** such as routers, switches, servers, and printers from a central location.

SNMP Components:

- **Manager (NMS – Network Management System):** A software application that runs on a central host and monitors the network. It sends requests and receives data from agents.
- **Agent:** Software running on each managed device. It collects local device information and reports it to the manager. It can also send unsolicited alerts (traps).
- **MIB (Management Information Base):** A structured database on each device that defines what information can be monitored and managed. It uses a tree-like hierarchy of **OIDs (Object Identifiers)**.

SNMP Operations:

Operation	Description
GET	Manager requests the value of a specific OID from an agent.
GET-NEXT	Retrieves the next OID value in the MIB tree.
SET	Manager sets/modifies a value on the agent.
TRAP	Agent sends an unsolicited alert to the manager (e.g., device failure).
INFORM	Like TRAP, but requires acknowledgement from the manager.

SNMP Versions:

- **SNMPv1:** Basic, no security (community strings in plaintext).
- **SNMPv2c:** Improved performance; still uses community strings.
- **SNMPv3:** Adds authentication and encryption — the current standard.

Port: SNMP uses **UDP port 161** for queries and **port 162** for traps.